Structural Equilibrium, Member Response, and Design Evidence

CIVIL-ENGINEERING · ce-structural · BoardPrep lesson v1

Official examination scope

PRC lists engineering mechanics, strength of materials, theory of structures, reinforced-concrete and prestressed members, steel members and connections, footings, construction-material testing, and governing codes. The examples here teach analysis principles; they do not provide current member capacities or code design limits.

Learning objectives

- Use equilibrium to check reactions for a simply supported beam.
- Relate load, section properties, stress, and deformation within stated assumptions.
- Trace a load path and distinguish analysis from code-based design.

1. Equilibrium before member formulas

A structure is in static equilibrium when the sum of forces and moments is zero in the chosen coordinate system. Draw a free-body diagram that isolates the member and shows applied actions, support reactions, distances, and sign convention. Solve independent equilibrium equations, then substitute the reactions back into them as a check. Symmetry can simplify a result only when geometry and loading are actually symmetric. A correct reaction calculation is a boundary condition for later shear and moment analysis; it does not establish that the member is adequately sized.

2. Stress, section properties, and deformation

For a prismatic elastic member under idealized axial load, average normal stress is P/A. In elementary elastic bending, the flexure relation = My/I connects bending moment, distance from the neutral axis, and section second moment of area. Deflection formulas require their stated support, load, geometry, and linear-elastic assumptions. Use consistent force and length units, and check whether the requested output is stress, force, or displacement. A formula is not a substitute for checking stability, shear, serviceability, connection behavior, or applicable design provisions.

3. Load path, structural systems, and materials

Loads move through connected members to supports and foundations. At each interface, identify the action transferred, the connection that carries it, and the next resisting element. Reinforced concrete relies on composite action between concrete and reinforcement; steel framing depends on member and connection behavior; prestressing introduces a deliberate force state that changes service response. Material tests provide measured evidence under a method and specimen condition; results should be interpreted with the test’s limits. Do not infer a full structural capacity from one material property or one isolated member calculation.

4. From analysis to design review

Design requires the applicable current code, load combinations, material specifications, detailing rules, and project conditions. A classroom analysis can expose an equilibrium error or explain a member response, but it cannot certify compliance. Track code edition and project jurisdiction, check assumptions against the drawings, and seek the responsible professional review when the model or loading is uncertain. Independent checks should target likely failure paths: units, sign convention, support conditions, section axis, load combinations, and whether a reported demand and capacity use compatible definitions.

Find the reactions of a symmetric beam

Question: A simply supported beam carries a 10 kN point load at midspan over a 6 m span. Find each support reaction.
1. Draw the free-body diagram with vertical reactions RA and RB.
2. Use vertical equilibrium: RA + RB - 10 = 0 kN.
3. Take moments about A: RB(6) - 10(3) = 0, so RB = 5 kN; then RA = 5 kN.
Answer: Each support reaction is 5 kN upward.
Check: The reactions sum to 10 kN and their moments balance about either support.

Common misconceptions

- A symmetric answer is justified when the loading or supports are not symmetric.
- Passing an equilibrium check proves the member is safe.
- A material test value is itself a design resistance.

Practice patterns

- Sketch the free body and sign convention before solving.
- Check force and moment balance independently.
- Name the code and assumptions needed before claiming a design result.

Official scope references

Official scope: Revised Program of the Civil Engineers Licensure Examination March 2026 — Professional Regulation Commission; Professional Regulatory Board of Civil Engineering — https://www.prc.gov.ph/sites/default/files/RevisedExamProgramMarch2026CivilEngg.pdf

Instructional references

Instructional reference: Analysis of Structural Member Systems — Jerome J. Connor; MIT OpenCourseWare — https://ocw.mit.edu/courses/1-571-structural-analysis-and-control-spring-2004/resources/open-textbooks/
Instructional reference: Structural Engineering Design Lecture Notes — Oral Buyukozturk; MIT OpenCourseWare — https://ocw.mit.edu/courses/1-051-structural-engineering-design-fall-2003/pages/lecture-notes/

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